OKA REPRESENTATIVES OF HOMOTOPY TYPES OF COUNTABLE FINITE-DIMENSIONAL CW COMPLEXES

PENG-CHAO WANG

Abstract. We construct an Oka manifold homotopy equivalent to every nonempty countable CW complex of finite dimension. For a complex of dimension d, the representative has complex dimension $5d+7$ and a countable good open cover by copies of $\mathbb{C}^{5d+7}$. For a finite simplicial complex with s maximal simplices, a separate construction gives a finite such cover in complex dimension $5s+2$, with precisely the prescribed nerve. The construction glues copies of complex Euclidean space by fiberwise holomorphic translations into a Hausdorff manifold with a good open cover, whose nerve determines its homotopy type by the nerve lemma.

1. Introduction

Which homotopy types admit Oka representatives? The realization question for finite CW complexes goes back to Gromov and is recorded in Forstnerič's survey [3, Problem 6.24]. We give a construction for finite complexes and, more generally, for countable CW complexes of finite dimension. The manifolds we build are covered by open subsets biholomorphic to complex Euclidean spaces, with contractible finite intersections.

We use the strong dominability criterion of Du, Guo, Wang, and Xie [1, Theorem 1.3]: a connected complex manifold is Oka if it is strongly dominable. Here a complex D-dimensional manifold M is *strongly dominable* if, for each $p \in M$, there is a holomorphic map $F : \mathbb{C}^D \to M$ with $F(0) = p$ and dF_0 an isomorphism. An entire Euclidean chart supplies such a map by translation. Our gluing construction produces strongly dominable manifolds, whose Oka property follows from this criterion.

All complex manifolds considered below are Hausdorff and second countable, but they need not be connected. For a disconnected manifold, Oka means that each connected component is Oka. We use the usual convex approximation characterization: a complex manifold Y is Oka if, for every $m \geq 1$, every compact convex set $K \subset \mathbb{C}^m$ and every map $f : U \to Y$ holomorphic on a neighbourhood U of K, the restriction $f|_K$ is a uniform limit of entire maps $\mathbb{C}^m \to Y$; see [3, Definition 1.1]. A *good open cover* is an open cover for which every nonempty finite intersection is contractible.

Theorem 1.1. *Let K be a nonempty countable CW complex of finite dimension d. There is an Oka manifold M of complex dimension $5d+7$ such that $M \simeq K$. Moreover, M admits a countable good open cover, every member of which is biholomorphic to $\mathbb{C}^{5d+7}$.*

The finite case also admits a model with a finite cover and a prescribed nerve. For a simplicial complex L, nonempty means $|L| \neq \varnothing$.

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Theorem 1.2. *Let L be a nonempty finite simplicial complex with v vertices and s maximal simplices. There is an Oka manifold M_L of complex dimension $D = 5s + 2$ with a good open cover $W_1, \dots, W_v$ such that each W_i is biholomorphic to $\mathbb{C}^D$ and the nerve of the cover is exactly L. In particular, $M_L \simeq |L|$.*

For finite complexes, Theorem 1.1 gives a bound depending only on dimension, whereas Theorem 1.2 provides a finite cover with the specified nerve. The finite models also have trivial canonical bundle and topologically trivial complex tangent bundle (Proposition 5.1).

Remark 1.3. The dimension bounds arise from Proposition 2.1. For convex domains in $\mathbb{C}^n$, Narasimhan's theorem supplies proper holomorphic embeddings into $\mathbb{C}^q$ with $q = 2n + 1$; duplicating their coordinates gives $\dim_{\mathbb{C}} M = n + 2q = 5n + 2$.

For the realization of a homotopy type alone, smaller representatives may exist. The Riemann sphere $\mathbb{C}P^1$ is complex homogeneous, hence Oka [3, Section 1], and is homotopy equivalent to S^2. It thus gives complex dimension 1, whereas Theorem 1.1 gives 17.

For a discrete complex with $v \geq 2$ vertices, the finite and countable constructions give dimensions $5v + 2$ and 7, respectively; for a d-simplex with $d \geq 1$, they give 7 and $5d + 7$. Thus neither numerical bound uniformly improves the other. In the discrete case, a disjoint union of v copies of $\mathbb{C}^7$ already has a finite good cover with the prescribed nerve.

For a compact complex manifold, every open cover admits a finite refinement by Stein spaces whose nonempty intersections are contractible [2, Appendix A, Lemma A.1]; convex realizations of a prescribed nerve are constructed in [7, Definition 3.1 and Theorem 3.3]; and the nerve lemma we use is [4, Corollary 4G.3]. What we need here is different: each cover member must be biholomorphic to all of $\mathbb{C}^D$, and the gluing must preserve the prescribed nerve while producing a Hausdorff manifold.

Section 2 proves Proposition 2.1. Section 3 realizes finite nerves and proves Theorem 1.2. Section 4 treats Euclidean open sets and proves Theorem 1.1. In Section 5, we show that the model of Theorem 1.2 can be chosen with trivial canonical bundle and topologically trivial complex tangent bundle, and we apply the results to fundamental groups.

2. A Hausdorff gluing construction

Given a finite or countable family of convex domains, the following proposition constructs a complex manifold with a good cover by entire Euclidean charts and with the same nerve. The main difficulty is to make the gluing Hausdorff, which is a condition at the boundary of the overlaps $U_i \cap U_j$: there, the difference of two proper holomorphic embeddings can remain bounded through cancellation. To treat this, we choose proper holomorphic embeddings h_i of the domains and repeat their coordinates in two blocks, setting $H_i = (h_i, \lambda_i h_i)$ with pairwise distinct scalars λ_i. The two blocks allow us to recover both h_i and h_j from $H_i - H_j$. Properness then makes the graphs of these differences closed in the ambient spaces and yields separation in the quotient.

Proposition 2.1. *Let $n \geq 1$, and let $\{U_i\}_{i \in J}$ be a nonempty finite or countable family of nonempty convex open subsets of $\mathbb{C}^n$. Put $D = 5n + 2$. There is a strongly dominable complex manifold M, with an open cover $\{W_i\}_{i \in J}$, such that*

1. *each W_i is biholomorphic to $\mathbb{C}^D$;*
2. *for every finite $I \subset J$ with $|I| \geq 2$, and any $i \in I$, the chart W_i identifies $W_I = \bigcap_{j \in I} W_j$ with*

$$U_I \times \mathbb{C}^{4n+2}, \qquad U_I = \bigcap_{j \in I} U_j;$$

(3) *the cover is good and has the same nerve as* $\{U_i\}_{i\in J}$.

Consequently M *is Oka by* [1, Theorem 1.3].

Proof. Each U_i is Stein. By Narasimhan's embedding theorem [6], choose a proper holomorphic embedding

$$h_i : U_i \hookrightarrow \mathbb{C}^q, \qquad q = 2n+1.$$

Choose pairwise distinct constants $\lambda_i \in \mathbb{C}$ and duplicate these embeddings by setting

$$H_i(z) = (h_i(z), \lambda_i h_i(z)) \in \mathbb{C}^{2q}.$$

For $i \neq j$, set $F_{ij} = H_i - H_j$ on $U_i \cap U_j$. The two coordinate blocks recover the original embeddings: if $F_{ij}(z) = (a, b)$, then

$$h_i(z) = \frac{b - \lambda_j a}{\lambda_i - \lambda_j}, \qquad h_j(z) = \frac{b - \lambda_i a}{\lambda_i - \lambda_j}. \tag{1}$$

These recovery formulas make F_{ij} proper. Indeed, the inverse image of a compact set is closed in $U_i \cap U_j$ and, by (1), lies in the intersection of two compact sets supplied by the properness of h_i and h_j. That intersection is a compact subset of $U_i \cap U_j$. In particular, the graph of F_{ij} is closed in $\mathbb{C}^n \times \mathbb{C}^{2q}$: a convergent sequence in the graph has bounded second coordinate, so its first coordinates remain in a compact subset of $U_i \cap U_j$. The limit remains in the overlap, and continuity identifies its second coordinate.

For the gluing, take the disjoint union of entire Euclidean spaces

$$V = \coprod_{i\in J} V_i, \qquad V_i = \mathbb{C}^n \times \mathbb{C}^{2q} \cong \mathbb{C}^D.$$

The overlaps $U_i \cap U_j$ specify where different copies are identified; each V_i itself is an entire Euclidean space. On V impose the following relation. Points in the same V_i are equivalent only when they are equal. For $i \neq j$, set

$$(z, w)_i \sim (z', w')_j \quad \Longleftrightarrow \quad z = z' \in U_i \cap U_j, \quad w' = w + H_i(z) - H_j(z). \tag{2}$$

This is already an equivalence relation. Symmetry follows by reversing the difference. For transitivity, the differences telescope, and membership in $U_i \cap U_j$ and $U_j \cap U_k$ implies membership in $U_i \cap U_k$. If $k = i$, the two differences cancel and the two points in V_i are equal. Hence no additional identifications arise from chains of transitions.

Let $M = V/\!\sim$ and let $Q : V \to M$ be the quotient map. The transition between V_i and V_j is the biholomorphism

$$\Phi_{ij}(z, w) = (z, w + H_i(z) - H_j(z)) \tag{3}$$

on $(U_i \cap U_j) \times \mathbb{C}^{2q}$. It fixes z and translates w by a holomorphic function of z. Its inverse is Φ_{ji}, and $\Phi_{jk} \circ \Phi_{ij} = \Phi_{ik}$ on triple overlaps. For an open set $O \subset V$, put $O_i = O \cap V_i$. Its saturation $Q^{-1}(Q(O))$ intersects V_j in

$$O_j \ \cup \bigcup_{i\neq j} \Phi_{ij}\big(O_i \cap ((U_i \cap U_j) \times \mathbb{C}^{2q})\big), \tag{4}$$

which is open. Thus Q is open.

The equivalence relation $R \subset V \times V$ is closed. On $V_i \times V_i$ it is the diagonal. For $i \neq j$, it is described by $z = z'$ and $(z, w' - w) \in \mathrm{Graph}(F_{ij})$, so closedness follows from the ambient graph closedness proved above. Since the blocks $V_i \times V_j$ form a disjoint open cover of $V \times V$, closedness on each block gives closedness of R, even when J is infinite.

An open quotient with closed equivalence relation is Hausdorff. To see this directly, choose representatives $x, y \in V$ of two different classes. Since $(x, y) \notin R$, there are open neighborhoods

$O \ni x$ and $O' \ni y$ with $(O \times O') \cap R = \varnothing$. The open sets $Q(O)$ and $Q(O')$ are disjoint neighborhoods of the two classes.

Clearly, the restriction $Q|_{V_i}$ is an open injection, hence a homeomorphism onto the open set $W_i = Q(V_i)$. The biholomorphic transitions (3) equip M with a complex manifold structure. The images under Q of the countable bases of the V_i obviously form a countable base for M. Hence M is second countable and paracompact.

For $|I| \geq 2$, the exact relation (2) identifies W_I in the ith chart with $U_I \times \mathbb{C}^{2q}$. For a singleton $I = \{i\}$, the intersection is the entire $W_i \cong \mathbb{C}^D$. Every nonempty U_I is convex, so all nonempty finite intersections of the W_i are contractible. Their nonemptiness is precisely the nonemptiness of the corresponding intersections of the U_i.

Finally, if $p \in W_i$ has coordinate $a \in \mathbb{C}^D$, the map $\zeta \mapsto Q((a+\zeta)_i)$ is entire, takes 0 to p, and has invertible differential there. Thus M is strongly dominable. Each connected component is Oka by [1, Theorem 1.3]. □

Remark 2.2. Since M is paracompact and the cover in Proposition 2.1 is good, the nerve lemma [4, Corollary 4G.3] gives $M \simeq |N(\{U_i\})|$.

3. Finite nerves and finite covers

We recall the convex nerve construction in [7, Definition 3.1 and the proof of Theorem 3.3], together with the passage to open sets by small neighborhoods used in the proof of [7, Theorem 3.14]. The proof below fixes the ambient complex dimension and the choice of a common thickening radius.

Lemma 3.1. *Let L be a nonempty finite simplicial complex with v vertices and s maximal simplices. There are bounded convex open sets $U_1, \ldots, U_v \subset \mathbb{C}^s$ whose nerve is L.*

Proof. Label the vertices $1, \ldots, v$, and write $\sigma_1, \ldots, \sigma_s$ for the maximal simplices. Let $e_1, \ldots, e_s$ be the standard basis vectors of $\mathbb{R}^s$, and set

$$C_i = \operatorname{conv}\{e_a : i \in \sigma_a\}.$$

These are nonempty compact faces of the standard simplex. Uniqueness of barycentric coordinates gives, for every nonempty vertex set I,

$$\bigcap_{i \in I} C_i = \operatorname{conv}\{e_a : I \subset \sigma_a\},$$

where the convex hull of the empty set is empty. The intersection is therefore nonempty exactly when I is a simplex of L.

Regard $\mathbb{R}^s$ as the standard real subspace of $\mathbb{C}^s$. For each nonface I, the continuous function

$$z \longmapsto \max_{i \in I} \operatorname{dist}(z, C_i)$$

tends to infinity as $|z| \to \infty$ and has a strictly positive minimum δ_I. There are finitely many nonfaces, so choose $\epsilon > 0$ smaller than every δ_I; if there are no nonfaces, choose any $\epsilon > 0$. The sets

$$U_i = \{z \in \mathbb{C}^s : \operatorname{dist}(z, C_i) < \epsilon\}$$

are bounded, convex, and open. Intersections corresponding to faces retain their points in the C_i, whereas intersections corresponding to nonfaces remain empty by the choice of ϵ. Thus their nerve is L. □

Proof of Theorem 1.2. Apply Lemma 3.1 and then Proposition 2.1 with $n = s$. The resulting cover has v members, its nerve is L, and its charts have complex dimension $5s+2$. Remark 2.2 gives $M_L \simeq |L|$. □

Corollary 3.2. *Every nonempty finite CW complex is homotopy equivalent to an Oka manifold admitting a finite good open cover by entire Euclidean charts.*

Proof. By [4, Theorem 2C.5], a finite CW complex is homotopy equivalent to a finite simplicial complex. Apply Theorem 1.2 to that complex. □

4. Euclidean open sets and countable CW complexes

We reduce the countable CW case to open subsets of complex Euclidean space, for which we first construct Oka representatives.

Proposition 4.1. *Let $U \subset \mathbb{C}^n$ be a nonempty open set, with $n \geq 1$. There is an Oka manifold M_U of complex dimension $5n + 2$ such that $M_U \simeq U$, with a countable good open cover by entire Euclidean charts.*

Proof. Choose a countable cover $\{U_i\}$ of U by Euclidean open balls contained in U. Every nonempty finite intersection is convex. Proposition 2.1 gives M_U with the same nerve as this ball cover. Both U and M_U are paracompact and both covers are good. The nerve lemma therefore gives

$$M_U \simeq |N(\{U_i\})| \simeq U.$$

Only the covering balls, rather than U itself, must be Stein for the embedding step in Proposition 2.1. □

Proof of Theorem 1.1. By the classical thickening construction in [5, Section 2, Fact 2 and its proof], there is a nonempty open set $V \subset \mathbb{R}^{2d+1}$ with $V \simeq K$. The cited proof first replaces K by a locally finite d-dimensional polyhedron and then takes the interior of a regular neighborhood of an embedding in $\mathbb{R}^{2d+1}$.

To pass to complex Euclidean space, set

$$U = V \times \mathbb{R} \subset \mathbb{R}^{2d+2} = \mathbb{C}^{d+1}.$$

This is a nonempty open subset of the complex Euclidean space, and projection to V is a homotopy equivalence. Applying Proposition 4.1 with $n = d + 1$ gives

$$M_U \simeq U \simeq V \simeq K, \qquad \dim_{\mathbb{C}} M_U = 5(d + 1) + 2 = 5d + 7.$$

The same proposition supplies the required countable good cover. The argument includes $d = 0$, since then $n = 1$. □

5. Bundles and fundamental groups

In this section, we use the fiberwise translations to strengthen Theorem 1.2. We also give a corollary on fundamental groups, with Oka realizations for countable groups and finite good covers in the finitely presented case.

Proposition 5.1. *The manifold M_L in Theorem 1.2 can be chosen with holomorphically trivial canonical bundle and topologically trivial complex tangent bundle.*

Proof. Use the model constructed in Proposition 2.1 with $n = s$ and $q = 2n+1$. Its transition differentials are

$$D\Phi_{ij} = \begin{pmatrix} I_n & 0 \\ dH_i - dH_j & I_{2q} \end{pmatrix}.$$

Their determinants are 1. Hence the standard holomorphic volume forms $dz_1 \wedge \cdots \wedge dz_n \wedge dw_1 \wedge \cdots \wedge dw_{2q}$ agree on overlaps and give a nowhere vanishing global section of the canonical bundle.

The unchanged z coordinates define a holomorphic submersion $p : M_L \to \mathbb{C}^n$. The vertical tangent frames also agree on overlaps, so there is an exact sequence of holomorphic vector bundles

$$0 \longrightarrow M_L \times \mathbb{C}^{2q} \longrightarrow TM_L \xrightarrow{dp} M_L \times \mathbb{C}^n \longrightarrow 0.$$

Because M_L is paracompact, a continuous Hermitian metric gives a topological complex bundle splitting by taking the orthogonal complement of the vertical subbundle. Both outer bundles are trivial, and thus $TM_L \cong M_L \times \mathbb{C}^{n+2q}$ as a topological complex vector bundle. No holomorphic splitting is required. □

Corollary 5.2. *Every countable group is the fundamental group of a connected Oka manifold of complex dimension* 17*. Every finitely presented group also admits an Oka representative of its finite presentation complex with a finite good cover by entire Euclidean charts.*

Proof. A countable group Γ has a countable presentation: use its elements as generators and the multiplication-table relations $x_g x_h = x_{gh}$. The associated presentation CW complex K_Γ is connected, countable, and two-dimensional. Applying Theorem 1.1 with $d = 2$ gives a connected Oka manifold M with

$$\dim_{\mathbb{C}} M = 17, \qquad \pi_1(M) \cong \Gamma.$$

For a finite presentation, its finite presentation complex is covered by Corollary 3.2. □

If Γ has a $K(\Gamma, 1)$ that is a CW complex with countably many cells and finite dimension, then Theorem 1.1 realizes that entire homotopy type, not only its fundamental group.

Statement on the use of artificial intelligence

Under the author's guidance, ChatGPT helped develop the overall approach, supplement relevant background knowledge, refine details, and revise the manuscript.

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Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: wangpengchao26@mails.ucas.ac.cn